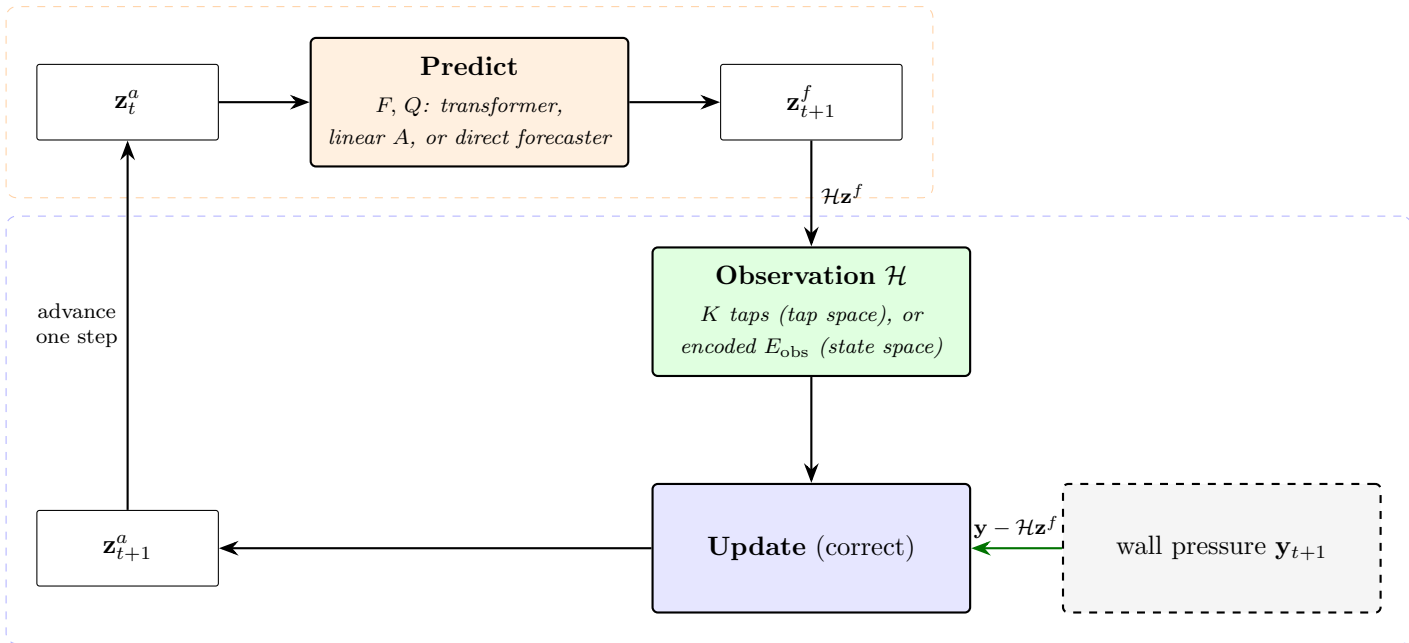


Predict: field free, no vorticity enters



Correct: sparse wall pressure

Production (two-stage): direct forecaster + encoded E_{obs} inside the impact window, matched transformer + tap space outside.